

Objective

The objective of this task is to develop a simple **Menu-Driven Bank Account Simulator** that allows users to perform common banking operations such as checking their balance, depositing money, withdrawing money, and viewing transaction history. The project helps practice Python functions, loops, validation, data structures, and state management while applying these concepts to a practical banking scenario. In the Streamlit version, session state is used to maintain the account balance and transaction history while the application is running.

Objectives

- Practice Python functions.
- Understand application state management.
- Implement deposit and withdrawal operations.
- Perform balance calculations.
- Maintain transaction history.
- Practice input validation.
- Prevent invalid withdrawals.
- Build an interactive Streamlit application.

3. Key Features

The application provides a simple banking interface where users can select different operations from the sidebar. Users can check their current balance, deposit money, withdraw money, and view their complete transaction history. The application validates transaction amounts and prevents withdrawals when the requested amount is greater than the available balance.

4. Technologies Used

The project uses **Python** for implementing the banking logic and **Streamlit** for creating the interactive user interface. Python functions are used to separate banking operations, while dictionaries and lists are used to store transaction information. Streamlit's session state maintains the account balance and transaction history while the application is running.

5. Deliverables

The completed project provides a working bank account simulator with the core banking operations described in the task.

Deliverables

- Bank account simulator
- Balance inquiry
- Deposit functionality
- Withdrawal functionality
- Transaction history
- Balance calculation
- Input validation
- Insufficient-balance protection
- Streamlit user interface

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY17>

```
app.py > deposit
1 import streamlit as st
2 st.set_page_config(
3     page_title="Bank Account Simulator",
4     page_icon="🏦",
5     layout="centered"
6 )
7
8 if "balance" not in st.session_state:
9     st.session_state.balance = 0.0
10
11 if "transactions" not in st.session_state:
12     st.session_state.transactions = []
13
14 def deposit(amount):
15     """Add money to the bank account."""
16     if amount <= 0:
17         return False, "Deposit amount must be greater than ₹0."
18
19     st.session_state.balance += amount
20
21     st.session_state.transactions.append({
22         "type": "Deposit",
23         "amount": amount,
24         "balance": st.session_state.balance
25     })
26
27     return True, f"₹{amount:.2f} deposited successfully."
28
29
30 def withdraw(amount):
31     """Withdraw money if sufficient balance is available."""
32     if amount <= 0:
33         return False, "Withdrawal amount must be greater than ₹0."
34
35     if amount > st.session_state.balance:
36         return False, "Insufficient balance. Withdrawal rejected."
37
38     st.session_state.balance -= amount
39
40     st.session_state.transactions.append({
41         "type": "Withdrawal",
42         "amount": amount,
43         "balance": st.session_state.balance
44
45
46 def withdraw(amount):
47     })
48     return True, f"₹{amount:.2f} withdrawn successfully."
49
50
51 def get_balance():
52     """Return current account balance."""
53     return st.session_state.balance
54
55 st.title("🏦 Bank Account Simulator")
56
57 st.write(
58     "A simple menu-driven banking application for "
59     "deposit, withdrawal, balance inquiry, and transaction history."
60 )
61
62 st.divider()
63
64 # Current balance
65 st.metric(
66     "Current Balance",
67     f"₹{get_balance():.2f}"
68 )
69
70 st.sidebar.title("🏦 Banking Menu")
71
72 menu = st.sidebar.radio(
73     "Select an operation:",
74     [
75         "Balance Inquiry",
76         "Deposit Money",
77         "Withdraw Money",
78         "Transaction History"
79     ]
80 )
81
82 if menu == "Balance Inquiry":
83
84     st.header("🏦 Balance Inquiry")
85
86     st.success(
87         f"Your current account balance is "
```

OUTPUT: _http://localhost:8501



Bank Account Simulator

A simple menu-driven banking application for deposit, withdrawal, balance inquiry, and transaction history.

Current Balance

₹0.00



Deposit Money

Enter deposit amount

100000.00

Deposit

₹100,000.00 deposited successfully.

New Balance: ₹100,000.00



Bank Account Simulator

A simple menu-driven banking application for deposit, withdrawal, balance inquiry, and transaction history.

Current Balance

₹100,000.00



Withdraw Money

Enter withdrawal amount

10000.00

Withdraw

₹10,000.00 withdrawn successfully.

Remaining Balance: ₹90,000.00



Bank Account Simulator

A simple menu-driven banking application for deposit, withdrawal, balance inquiry, and transaction history.

Current Balance

₹5,000.00



Balance Inquiry

Your current account balance is ₹5,000.00



Bank Account Simulator

A simple menu-driven banking application for deposit, withdrawal, balance inquiry, and transaction history.

Current Balance

₹90,000.00



Transaction History

1. Deposit

Amount: ₹100,000.00

Balance After Transaction: ₹100,000.00

2. Withdrawal

Amount: ₹10,000.00

Balance After Transaction: ₹90,000.00

LinkedIn:- <https://lnkd.in/p/dsTyfU5U>



Bank Account Simulator

A simple menu-driven banking application for deposit, withdrawal, balance inquiry, and transaction history.



Current Balance

₹5,000.00



Balance Inquiry

Your current account balance is ₹5,000.00



Shambhavi E L • You
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Just now • 🌐



VEDA Internship – Python Programming Track | Day 17 Successfully completed Day 17 of my Python Programming Internship journey! Today, I developed a Menu-Driven Bank Account Simulator using Python and Streamlit. 🚀 The application supports essential... more



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GIF



Github:-<https://github.com/shambhavi0804/VEDA-internship/commit/b411edf8fbb09f7cc9a2b248d563804e971b0b78>

